Introduction to Game Theory: Static Games

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What is Game Theory?

The Basic Idea

Game theory studies what happens when **self-interested agents interact**.

“Self-interested” doesn’t mean selfish. It means each person has their own preferences about how things turn out, and acts on those preferences.

The big question: when your outcome depends on what *other people* do, how should you act?

Game theory turns up everywhere:

- **Ethics**: Is morality a solution to a coordination problem?
- **Political philosophy**: Why do we need governments?
- **Philosophy of science**: Why do scientists share results?
- **Social epistemology**: When should we defer to experts?

And it gives us precise tools for thinking about problems we've already been discussing—like Schelling's focal points.

Normal-Form Games

The Payoff Matrix

The simplest way to represent a game is with a **payoff matrix** (also called the "normal form").

Table 1: The Prisoner's Dilemma

	C	D
C	3, 3	0, 5
D	5, 0	1, 1

- Rows = Player 1's options
- Columns = Player 2's options
- Each cell has two numbers: **Player 1's payoff, Player 2's payoff**

	C	D
C	3, 3	0, 5
D	5, 0	1, 1

If Player 1 picks C and Player 2 picks D:

- Player 1 gets **0**
- Player 2 gets **5**

Higher numbers are better. These could represent anything: money, happiness, years not in prison, etc.

What You Need for a Game

Three ingredients:

1. **Players**: Who is involved? (At least two.)
2. **Actions**: What can each player do?
3. **Payoffs**: What does each player get for each combination of actions?

The payoff matrix captures all three at once.

Classic Games

The Prisoner's Dilemma

Two suspects are arrested and interrogated separately.

- If both **cooperate** (stay silent): light sentence for each.
- If both **defect** (confess): moderate sentence for each.
- If one defects and the other cooperates: the defector goes free, the cooperator gets a harsh sentence.

	Cooperate	Defect
Cooperate	3, 3	0, 5
Defect	5, 0	1, 1

What Should You Do?

	Cooperate	Defect
Cooperate	3, 3	0, 5
Defect	5, 0	1, 1

Think about it from Player 1's perspective:

- If Player 2 cooperates: I get 3 by cooperating, **5 by defecting**
- If Player 2 defects: I get 0 by cooperating, **1 by defecting**

No matter what Player 2 does, defecting is better for me. The same logic applies to Player 2. So both defect—even though both cooperating would be better for everyone.

The Dilemma

This is why it's called a *dilemma*:

- Individual rationality leads to (Defect, Defect) with payoffs (1, 1)
- But (Cooperate, Cooperate) with payoffs (3, 3) is better for *both* players

This tension between individual and collective rationality is one of the deepest problems in social philosophy.

We'll come back to this when we discuss **iterated** Prisoner's Dilemma and Axelrod's tournaments.

Not all games involve conflict. Sometimes the challenge is just to **coordinate**.

Table 5: Which side of the road?

	Left	Right
Left	1, 1	0, 0
Right	0, 0	1, 1

Neither player wants to mismatch. They just need to pick the same side. This is a **pure coordination game**: both players always get the same payoff.

This is basically the Schelling coordination problem we already studied.

Battle of the Sexes

What if you want to coordinate, but you *disagree about where*?

Table 6: Battle of the Sexes

	Movie A	Movie B
Movie A	2, 1	0, 0
Movie B	0, 0	1, 2

- Both prefer going together over going separately
- But Player 1 prefers Movie A, and Player 2 prefers Movie B
- This has elements of both **coordination** and **competition**

At the opposite extreme from coordination: **zero-sum games**.

Whatever one player gains, the other loses.

Table 7: Matching Pennies

	Heads	Tails
Heads	1, -1	-1, 1
Tails	-1, 1	1, -1

Player 1 wins if the coins match; Player 2 wins if they differ. This is pure competition—no room for cooperation.

The Spectrum

These games lie on a spectrum:

Pure coordination $\longleftrightarrow$ Mixed motives $\longleftrightarrow$ Pure competition

Coordination Game $\longleftrightarrow$ Battle of the Sexes $\longleftrightarrow$ Matching Pennies

Most real-world interactions are somewhere in the middle. Even bitter rivals usually have *some* shared interests (e.g., not destroying the world).

Prisoner's Dilemma vs. Stag Hunt

There's a game that *looks* like the Prisoner's Dilemma but works very differently.

Table 8: Stag Hunt

	Cooperate	Defect
Cooperate	5, 5	0, 3
Defect	3, 0	3, 3

The story: Two hunters can hunt a stag together (big payoff) or each hunt a hare alone (small but guaranteed payoff). Hunting stag only works if *both* commit to it.

What's Different?

Prisoner's Dilemma: Defecting is always better for you, regardless of what the other player does.

Stag Hunt: The best response *depends on what the other player does*.

- If they cooperate: I should cooperate too (5 > 3)
- If they defect: I should defect too (3 > 0)

In the Stag Hunt, the problem isn't that rationality *forces* you to defect. The problem is **trust**—will the other person hold up their end?

Why This Matters

When we model a real situation as a game, it matters *a lot* whether we think it's a Prisoner's Dilemma or a Stag Hunt.

- **PD framing**: People are tempted to free-ride. Solution: change the incentives (punishment, regulation).
- **Stag Hunt framing**: People would cooperate *if they trusted each other*. Solution: build trust, communicate, establish conventions.

Many debates in political philosophy (Hobbes vs. Rousseau, for instance) can be read as disagreements about which game best models social life.

Discussion

Think of a real-world situation that involves two or more people making choices that affect each other.

1. Is it more like a Prisoner's Dilemma or a Stag Hunt?
2. What turns on the answer?
3. Could you change the situation to shift it from one to the other?

Analyzing Games

Pareto Optimality

Before we ask "what will rational players do?", let's ask: "which outcomes are *good*?"

An outcome is **Pareto optimal** if there's no other outcome that makes *someone* better off without making *anyone* worse off.

- In the Prisoner's Dilemma, (Cooperate, Cooperate) is Pareto optimal
- (Defect, Defect) is **not** Pareto optimal—both players could do better

But Pareto optimality doesn't tell us what players *will* do. For that we need equilibrium concepts.

Here's the key idea behind Nash equilibrium.

Your **best response** to the other player's strategy is the action that gives you the highest payoff, *given what they're doing*.

In the Prisoner's Dilemma (bold = Player 1's best response):

	Cooperate	Defect
Cooperate	3, 3	0, 5
Defect	**5**, 0	**1**, 1

Now add Player 2's best responses too:

	Cooperate	Defect
Cooperate	3, 3	0, **5**
Defect	**5**, 0	**1**, **1**

Cells where *both* numbers are bold are Nash equilibria. Only (Defect, Defect) qualifies. Defect is the best response no matter what—that's what makes it a **dominant strategy**.

Nash Equilibrium

A **Nash equilibrium** is a combination of strategies where *every* player is playing a best response to everyone else.

In other words: **nobody wants to change**, given what everyone else is doing.

- Prisoner's Dilemma: (Defect, Defect) is the unique Nash equilibrium
- Coordination Game: (Left, Left) and (Right, Right) are both Nash equilibria
- Battle of the Sexes: (A, A) and (B, B) are both Nash equilibria

Finding Nash Equilibria

Here's a quick method for 2-player games. For each cell:

1. Check if Player 1's payoff is the best in its **column** (i.e., best response to that column choice)
2. Check if Player 2's payoff is the best in its **row** (i.e., best response to that row choice)
3. If **both** conditions hold, it's a Nash equilibrium

	Cooperate	Defect
Cooperate	3, 3	0, **5**
Defect	**5**, 0	**1**, **1**

The bold numbers are best responses. Both are bold in (Defect, Defect)—that's the Nash equilibrium.

Find the Nash equilibria:

	Left	Right
Up	2, 1	0, 0
Down	0, 0	1, 2

- (Up, Left): Player 1 gets 2 (best in column), Player 2 gets 1 (best in row). **Nash equilibrium**
- (Down, Right): Player 1 gets 1 (best in column), Player 2 gets 2 (best in row). **Nash equilibrium**

This game has two equilibria. Which one gets played? That's where focal points come back in. In the gendered examples that are often used to illustrate the game, the focal points are created and sustained by **social norms**.

Dominated Strategies

What’s a Dominated Strategy?

A strategy is **dominated** if there’s another strategy that is *always* at least as good, and *sometimes* strictly better.

If you have a dominated strategy, you should never play it. No matter what the other players do, you have something better available.

In the Prisoner’s Dilemma, Cooperate is dominated by Defect.

Two trucks each choose a spot (1–5) on a beach with 10 customers at each spot.

- Customers go to the nearest truck; ties split evenly

	1	2	3	4	5
1	25, 25	10, 40	15, 35	20, 30	25, 25
2	40, 10	25, 25	20, 30	25, 25	30, 20
3	35, 15	30, 20	25, 25	30, 20	35, 15
4	30, 20	25, 25	20, 30	25, 25	40, 10
5	25, 25	20, 30	15, 35	10, 40	25, 25

Step 1: Is spot 1 ever a good choice?

Compare spot 1 vs. spot 2 for Truck 1: no matter where the other truck parks, spot 2 always gets more customers.

Spot 1 is **dominated** by spot 2. By symmetry, spot 5 is dominated by spot 4. Eliminate both.

After removing spots 1 and 5:

	2	3	4
2	25, 25	20, 30	25, 25
3	30, 20	25, 25	30, 20
4	25, 25	20, 30	25, 25

Step 2: Now spot 2 is dominated by spot 3 (and spot 4 by spot 3). Eliminate both.

The unique solution: both trucks park at **spot 3**.

	3
3	25, 25

This is the **median voter theorem** in action: rational competitors both converge on the center.

Why This Works

Iterated elimination of dominated strategies works because:

1. A rational player would never play a dominated strategy
2. If you *know* the other player is rational, you know they won't play dominated strategies
3. Once you remove those, new strategies might become dominated
4. If you know the other player knows you're rational, you can eliminate those too...

This is essentially **common knowledge of rationality** at work. Each round of elimination requires one more level of "I know that you know that I know..."

Dominance vs. Nash Equilibrium

How do these concepts relate?

- Every Nash equilibrium survives iterated elimination of dominated strategies
- But not every strategy that survives is part of a Nash equilibrium
- Dominance is a **weaker** requirement: it rules out fewer things

Think of it as two levels of analysis:

1. **Dominated strategies**: “Don’t play anything stupid”
2. **Nash equilibrium**: “Play a best response to what others are doing”

Looking Ahead

What We've Covered

- What a game is: players, actions, payoffs
- Classic games: Prisoner's Dilemma, Coordination, Battle of the Sexes, Matching Pennies, Stag Hunt
- Why PD vs. Stag Hunt matters for modeling real situations
- Solution concepts: Pareto optimality, best response, Nash equilibrium
- Dominated strategies and iterated elimination

- **Dynamic games**: What changes when players move in sequence?
- **Backward induction**: Solving games by thinking ahead from the end
- **Subgame perfection**: Why some Nash equilibria involve empty threats

And then we'll look at Axelrod's famous tournaments on the **iterated Prisoner's Dilemma**: what happens when you play the same game against the same person over and over?

For Next Time

Read chapters 4 and 5 of Leyton-Brown and Shoham, on extensive-form games.

Think about this question:

> *You're playing chess. You can see that if your opponent plays optimally from this point, they'll win. But your opponent has been making mistakes. Should you play as if they'll be rational from here on, or try to exploit their mistakes?*